

Scientific Methods

AirOne ships **37 deterministic scientific analyses** across **9 categories**, implemented in `src/scientific/` and registered in `src/scientific/registry.py`. Every analysis returns a `ScientificResult` with a value, unit, explicit **uncertainty**, and the assumptions it relied on.

Scientific-honesty rules

These rules are enforced in the analysis code and are non-negotiable:

1. **Correlation is never reported as causation.** Correlation analyses (e.g. `radiation_altitude_correlation`, `correlation_matrix`) report the coefficient and sample size and explicitly label the relationship as correlation only.
2. **Uncertainty is always propagated.** No analysis returns a point value without an uncertainty (or an explicit statement that uncertainty is not yet quantifiable, in which case the result is marked accordingly).
3. **Insufficient data yields an explicit “cannot compute”, never a guess.** An analysis with too few valid samples returns an unavailable/insufficient result rather than a fabricated number.
4. **Invalid inputs are excluded, not imputed.** Analyses consume only valid measurements; missing values are never zero-filled.
5. **Simulated inputs stay labelled.** Results computed from `SIMULATED` data inherit that provenance.

Categories and analyses

Atmospheric (10)

`atmospheric_density`, `atmospheric_stability`, `boundary_layer_height`, `dew_point`, `hypsonetric_altitude`, `lapse_rate`, `pressure_gradient`, `temperature_gradient`, `temperature_inversion`, `vertical_profile`.

Core relations used include the barometric/hypsometric equation for pressure-derived altitude, the environmental lapse rate (dT/dz), the Magnus/Arden-Buck formula for dew point, and the ideal-gas relation for air density. Stability and inversion detection compare the measured lapse rate to the dry adiabatic lapse rate.

Descent (4)

`descent_rate`, `drag_coefficient`, `landing_detection`, `terminal_velocity`.

Descent rate is the time derivative of altitude; terminal velocity and the effective drag coefficient are derived from the balance of gravity and drag during steady descent. Landing detection combines low vertical speed with altitude stability.

Gases (3)

air_quality_proxy, plume_detection, voc_profile.

VOC/gas-resistance readings from the BME688 are turned into an air-quality proxy and a vertical VOC profile; plume detection flags localised concentration anomalies.

GNSS (4)

displacement_from_start, ground_speed, ground_track_distance, vertical_speed.

Horizontal metrics use great-circle (haversine) distance between fixes; vertical speed uses GNSS altitude deltas. All require valid GNSS fixes — with no fix, the analyses report unavailable rather than emitting zeros.

Magnetic (4)

magnetic_anomaly, magnetic_field_magnitude, magnetic_heading, magnetic_inclination.

Field magnitude is the Euclidean norm of the three magnetometer axes; heading and inclination are derived from the field vector; anomalies are deviations from the expected local field.

Power (3)

battery_health, brownout_risk, voltage_trend.

Voltage trend is a regression over the battery-voltage series; brownout risk compares projected voltage against a threshold; battery health summarises the discharge behaviour.

Radiation (4)

dose_rate, mean_count_rate, radiation_altitude_correlation, radiation_burst.

Count rates (CPM) are converted to an approximate dose rate; the altitude-radiation relationship is reported strictly as a correlation with its coefficient and sample count; bursts are transient rate excursions.

UV / Solar (2)

uv_altitude_trend, uv_index.

UV index from the UV sensor and its trend with altitude.

Composite (3)

change_rate_monitor, correlation_matrix, environmental_hazard_index.

Cross-sensor summaries: a rate-of-change monitor, a correlation matrix across key channels (correlation only — never causal), and a composite environmental hazard index.

Filtering and fusion

Before analyses run, the pipeline can apply configurable filters (stage 14) and sensor fusion (stage 15):

- **Kalman 1-D** (e.g. fused pressure), **moving average** (temperature), **EKF altitude** — all configured under pipeline.filter_config.
- Fused outputs are tagged DataSource.FUSED; filtered outputs are tagged appropriately so consumers always know the provenance of a number.

Narrative

GET /api/v1/narrative produces a natural-language description of the mission from the computed results. The narrative is generated from real analysis outputs and states uncertainty; it never asserts findings the analyses did not support.